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In [1]: #EXP 4_ MISSING VALUE TREATEMENT
```

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In [2]: #AIM: MISSING VALUE TREATEMENT
```

```
In [3]: # Exp_4
# Name:Suhani Milind Gayakwad
# Roll no:16
# Sec:A
# Subject:ET2
# Date:19/08/2025
```

```
In [4]: import pandas as pd
```

```
In [5]: import os
```

```
In [6]: os.getcwd()
```

```
Out[6]: 'C:\\Users\\user'
```

```
In [7]: os.chdir("C:\\Users\\user\\Desktop")
```

```
In [8]: data=pd.read_csv("titanic.csv")
```

```
In [9]: data.head(5)
```

```
Out[9]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	S
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
In [10]: data.size
```

Out[10]: 10692

```
In [11]: data.columns
```

Out[11]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'], dtype='object')

```
In [12]: data.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [13]: data.isna()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	False	False	False	False	False	False	False	False	False	False	True	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	True	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	True	False
...	...	...	...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False	False	True	False
887	False	False	False	False	False	False	False	False	False	False	False	False
888	False	False	False	False	False	True	False	False	False	False	True	False
889	False	False	False	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False	False	True	False

891 rows × 12 columns



```
In [14]: data.isna()
```

```
Out[14]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0		False	False	False	False	False	False	False	False	False	True	False
1		False	False	False	False	False	False	False	False	False	False	False
2		False	False	False	False	False	False	False	False	False	True	False
3		False	False	False	False	False	False	False	False	False	False	False
4		False	False	False	False	False	False	False	False	False	True	False
...	...	...	...	...	...	...	...	...	...	...	...	...
886		False	False	False	False	False	False	False	False	False	True	False
887		False	False	False	False	False	False	False	False	False	False	False
888		False	False	False	False	True	False	False	False	False	True	False
889		False	False	False	False	False	False	False	False	False	False	False
890		False	False	False	False	False	False	False	False	False	True	False

891 rows × 12 columns



```
In [15]: data.isna().any()
```

```
Out[15]: PassengerId    False
Survived      False
Pclass        False
Name          False
Sex           False
Age           True
SibSp         False
Parch         False
Ticket        False
Fare          False
Cabin         True
Embarked      True
dtype: bool
```

```
In [16]: df1=data.dropna()
```

```
In [17]: df1
```

```
Out[17]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46
10	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000	G6
11	12	1	1	Bonnell, Miss. Elizabeth	female	58.0	0	0	113783	26.5500	C103
...	...	...	...	...	...	...	...	...	...	...	...
871	872	1	1	Beckwith, Mrs. Richard Leonard (Sallie Monypeny)	female	47.0	1	1	11751	52.5542	D35
872	873	0	1	Carlsson, Mr. Frans Olof	male	33.0	0	0	695	5.0000	B51 B53 B55
879	880	1	1	Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)	female	56.0	0	1	11767	83.1583	C50
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148

In [18]:

Out[18]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
888		False	False	False	False	True	False	False	False	False	True	False
889		False	False	False	False	False	False	False	False	False	False	False
890		False	False	False	False	False	False	False	False	False	True	False

891 rows × 12 columns

In [19]: `df1.isna().any()`

Out[19]:

PassengerId	False
Survived	False
Pclass	False
Name	False
Sex	False
Age	False
SibSp	False
Parch	False
Ticket	False
Fare	False
Cabin	False
Embarked	False

dtype: bool

In [20]: `data["Age"].fillna(29.699118)`

Out[20]:

0	22.000000
1	38.000000
2	26.000000
3	35.000000
4	35.000000
...	
886	27.000000
887	19.000000
888	29.699118
889	26.000000
890	32.000000

Name: Age, Length: 891, dtype: float64

In []:

In []: